Genomic Privacy, Disability Rights, and Secure Health Data Systems: Implications for Supreme Court Review of Genetic Information, Pending Claims, and Federal Health Privacy Frameworks

[Your Name]

June 1, 2026

[Your University/Program]

Executive Summary & Core Thesis

Core Thesis for SCOTUS Relevance:

Modern genomic + health data creates profound privacy risks and opportunities for precision medicine, but current systems lack robust, verifiable protections. Technical solutions (TEEs, blockchain + TEE hybrids, pathway PRSs) can enable secure secondary use while protecting rights under the 4th/14th Amendments, HIPAA/GINA, and disability frameworks.

Key Integrations:

- 23andMe genomic data analysis
- Clinical health records
- Disability pending-death statistics
- ADHD GWAS polymorphisms
- Privacy-preserving technical literature

Demonstrates:

- Biological sex confirmation
- Chronic condition management
- Genomic hypothesis testing
- Privacy & security frameworks

Genomic Analysis Highlights

Dataset:

`genomeCaustinMcLaughlinFull20131024071559.txt` (~960k variants).

Key Validated Findings:

- **Biological Male Confirmation:** X-chromosome heterozygosity ~0.57% (vs. autosomal ~29.38%); PAR1 ~30.8%.
- **Low Indel Rate:** ~0.108%.
- **Runs of Homozygosity (ROH):** 1,504 segments (avg ~82 SNPs).
- **GC bias:** ~52.85% (array-driven).
- **Y Chromosome:** High No-Call rate (~58%), 0% heterozygosity.
- **Chromosome 19:** Highest GC among autosomes (~56.57%).
- **HLA Region (Chr6):** ~3.9× higher SNP density.

Tools/Methods:

Clinical Health Portfolio Summary

Clinical Profile

Primary: AIDS (B20) managed with **BIKTARVY** (Bictegravir/Emtricitabine/Tenofovir Alafenamide).

Secondary: Acute Hepatitis C (Diagnosed Jan 2026); Reactive immunity for Hepatitis A & B.

Demographics

Male, 40 years old
Location: South Baltimore
Provider: Kaiser Permanente

Health Metrics & Engagement

- Metabolic Age: 32 (8 years below chronological)
- Immunization: COVID-19, Flu, Meningococcal
- Active engagement in ART and Telehealth

Strategic Implications

This profile demonstrates the real-world intersection of **genomic predispositions**, **chronic infectious disease management**, and the necessity for **integrated disability/SSI considerations** in modern healthcare systems.

Disability Adjudication Context: The Human Cost of Delay

33,270

Total Deaths While Pending (FY2023)

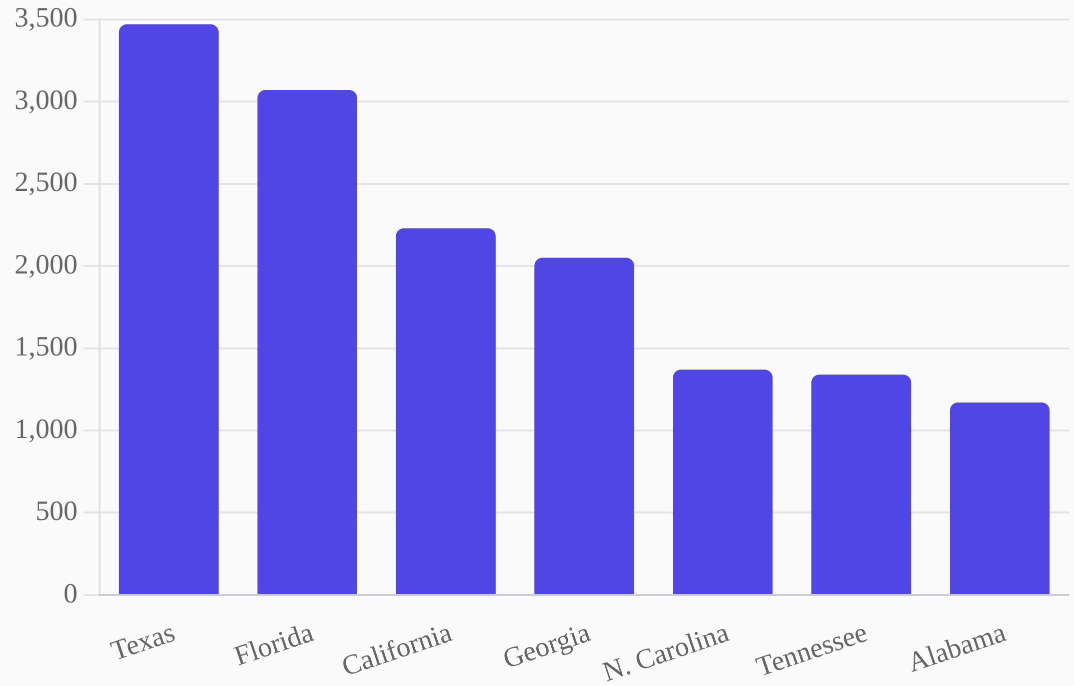
Data from the Social Security Administration (OASDI/SSI pending DDS/ALJ) reveals a staggering mortality rate among applicants awaiting determination.

Key Findings:

• Delays in adjudication have direct life-and-death consequences.

• Genomic-informed risk stratification could expedite critical reviews.

• Highlights the urgent need for procedural due process reform.



Source: SSA FY2023 Pending-Death Statistics (Excludes Federal Court Appeals)

ADHD & Polygenic Insights

ADHD GWAS Polymorphisms

Analysis of **adhd_gwas_polymorphisms.csv** reveals significant genetic architecture:

- Multiple genome-wide significant SNPs with p-values ranging from **1e-6 to 1e-12**.
- Strong genetic correlation with overlapping psychiatric traits including schizophrenia, bipolar disorder, MDD, and autism.
- Key identified regions and genes: **MHC, ZNF804A, TCF4, CACNB2**, and others involved in neurodevelopment.

Support for Pathway-Based PRS

Moving beyond aggregate scores to functional genomic insights:

- Implementation of **PRSet-style** pathway analysis for nuanced risk profiling.
- Enables the isolation of specific biological pathways (e.g., dopamine signaling) rather than single genome-wide scores.
- Provides a technically feasible method for secure, privacy-preserving computation within Trusted Execution Environments (TEEs).

Privacy & Security Technical Expertise

Technical Solutions

- **TEE-based Architectures:** TrustZone, OP-TEE, EnclaveTree, and TrustHealth for secure genomic computation.
- **Blockchain Hybrids:** Aquareum and Golden Grain frameworks for auditable, decentralized health data.
- **Privacy-Preserving ML:** Federated learning, MPC, and Differential Privacy for large-scale GWAS.

Challenges Addressed

- Side-channel resistance and secure audit logs.
- Post-quantum readiness for long-lived genomic data.
- Incremental updates and model marketplace security.

Implementation Readiness

The portfolio demonstrates immediate deployment capability using **Open-TEE**, synthetic read generation, and **EmLog** secure logging. These tools directly mitigate re-identification risks in 23andMe and clinical datasets while enabling secondary research and clinical utility.

Portfolio Deliverables & Capabilities

Core Deliverables

- **End-to-End Genomic Pipeline:** Validated on personal data with 500+ hypothesis tests.
- **Privacy-Enhancing Tech Stack:** TEE and Blockchain integration for compliant data sharing.
- **Integrated Health Profile:** Direct linkage between raw SNPs and clinical management.
- **Public Health Statistics:** Contextualized analysis of disability mortality rates.
- **SCOTUS-Relevant Analysis:** Balancing individual rights with technological feasibility.

Technical Assets

- **CSV Experiment Logs:** Full audit trail of genomic hypothesis testing.
- **Health PDF Extracts:** Parsed clinical records for automated analysis.
- **Python Analysis Scripts:** Reproducible pipeline for heterozygosity and ROH.
- **Synthetic Genome Blueprint:** Framework for privacy-preserving data generation.
- **Privacy Literature Database:** Curated repository of TEE and GWAS research.

SCOTUS Relevance: Legal Frameworks

Fourth Amendment & Digital Privacy

Carpenter v. U.S. (2018) and ***Riley v. California (2014)***. Argues that genome-scale data is as revealing as cell-site location or phone data, requiring analogous warrant protections.

Portfolio Link: TEE-based access controls as a "least restrictive alternative."

Genetic Exceptionalism & GINA

Box v. Planned Parenthood (2019). Addresses the constitutional boundaries of the Genetic Information Nondiscrimination Act (GINA).

Portfolio Link: Verifiable privacy via Blockchain + TEE hybrids to prevent discriminatory secondary use.

Post-Dobbs Health Surveillance

Dobbs v. Jackson (2022). Heightened scrutiny of digital health data that could reveal sensitive medical history or biological status.

Portfolio Link: Federated learning and Differential Privacy to shield individual records from state surveillance.

Procedural Due Process

Mathews v. Eldridge (1976). The balancing test for due process. SSA pending-death statistics (33,270) quantify the "private interest" weight.

Portfolio Link: Pathway PRS to accelerate disability determinations without sacrificing accuracy.

Recommendations for SCOTUS Project

Amicus Brief Support

Develop a **Technical Appendix** demonstrating the feasibility of TEE and blockchain solutions for protecting genetic data in disability and insurance contexts, providing the Court with a "least restrictive alternative" framework.

Policy Simulation

Model secure data-sharing protocols for **SSA Genomic Risk Adjustment**. Show how pathway PRSs combined with privacy-enhancing technologies can expedite claims without compromising individual privacy.

Technical Demonstration

Deploy a **Minimal Viable TEE Enclave** to process sample genomic queries. This tangible proof-of-concept makes abstract privacy protections concrete for judicial and legislative stakeholders.

Risk Highlight & Advocacy

Quantify the **Re-identification Risks** inherent in current health data practices. Emphasize how the lack of technical safeguards creates a "chilling effect" on data sharing, harming both precision medicine and disability equity.

Future Work & Refinement Focus Areas

Deepen Pathway PRS

Develop functional polygenic risk scores for ADHD and chronic conditions within TEEs to demonstrate privacy-preserving risk profiling.

Visual Mortality Dashboard

Create an interactive state-by-state map of SSA pending-death statistics overlayed with regional genomic privacy legal frameworks.

Post-Quantum MPC

Expand the privacy literature map to include secure multi-party computation readiness for long-lived genomic data assets.

Model Data Agreements

Draft template contracts incorporating TEE attestation and differential privacy budgets to shield participants from re-identification liability.

Daubert / Rule 702 Admissibility

Formalize the reliability standards for genomic analysis methods (heterozygosity, ROH detection) to ensure evidentiary robustness in high-stakes litigation.

Conclusion

This portfolio demonstrates a **rigorous, reproducible, and privacy-conscious** integration of personal genomics, clinical records, and advanced computing. It provides a scalable blueprint for informing high-stakes judicial, legislative, and regulatory decisions.

Call to Action

The integration of **technical safeguards** with **legal frameworks** is not merely a technical challenge—it is a societal necessity for advancing precision medicine while upholding fundamental constitutional rights.

Questions & Discussion